

Scripts SQL — schéma PostgreSQL de la refonte

Scripts de création de la nouvelle structure de données du CRM LiVrai. Exécutés et vérifiés sur PostgreSQL 18.

Projet Refonte CRM LiVrai SGBD cible PostgreSQL 18 Fichiers db/postgres/ Septembre 2026

Contenu de l'annexe

FICHIER	PRODUCTION	RÔLE
01-schema.sql	Oui	Trois types énumérés, sept tables, contraintes d'intégrité et de validation, neuf index, déclencheurs, séquences de références métier.
02-seed.sql	Non	Jeu de données de démonstration pour le développement et la recette.

CORRESPONDANCE AVEC LES CONSTATS D'AUDIT

CONSTAT D'AUDIT	TRAITEMENT DANS LE SCHÉMA
D-05 — mots de passe en clair	password_hash, empreinte BCrypt
D-09 — autorisation par booléen admin	Type user_role à quatre valeurs
D-10 — volume -99 accepté et facturable	CHECK (volume_m3 > 0), CHECK (weight_kg > 0)
D-14 — statuts en texte libre français	Types delivery_status et invoice_status
D-04 — aucun index sur status	idx_delivery_status_created, index partiel idx_delivery_pending
Absence de traçabilité des actions	Table delivery_status_history

02-seed.sql ne doit jamais être exécuté en production. Les mots de passe des comptes de démonstration y sont publics (Livrai2026!). En production, le compte d'administration initial est créé au démarrage applicatif, avec un secret fourni par l'environnement.

Choix de conception assumé. invoice_line ne porte aucune clé étrangère vers delivery : la livraison facturée est désignée en clair dans le libellé. Ce choix supprime tout chemin circulaire du schéma, mais retire au SGBD la capacité d'empêcher une double facturation — ce contrôle relève désormais de la couche service. Voir ADR-11 du document d'architecture.

Lecture allégée. Les commentaires SQL ont été retirés de cette annexe pour en faciliter la lecture. Les fichiers .sql du dépôt les conservent intégralement : ils documentent chaque choix et son rattachement aux constats de l'audit.

EXÉCUTION

Ligne de commande

```
docker compose up -d db
```

```
psql -U livrai -d livrai -f 01-schema.sql
```

```
psql -U livrai -d livrai -f 02-seed.sql
```

01-schema.sql

Structure de la base. Seul script destiné à la production.

db/postgres/01-schema.sql PostgreSQL 18 216 lignes

```

1  CREATE TYPE delivery_status AS ENUM (
2      'PENDING',
3      'ACCEPTED',
4      'REJECTED',
5      'IN_TRANSIT',
6      'DELIVERED',
7      'BILLED',
8      'CANCELLED'
9  );
10
11 CREATE TYPE user_role AS ENUM (
12     'CLIENT',
13     'COMMERCIAL',
14     'EXPLOITATION',
15     'ADMIN'
16 );
17
18 CREATE TYPE invoice_status AS ENUM (
19     'DRAFT',
20     'ISSUED',
21     'PAID',
22     'CANCELLED'
23 );
24
25 CREATE TABLE app_user (
26     id                BIGINT          GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
27     email              VARCHAR(255) NOT NULL,
28     password_hash     VARCHAR(255) NOT NULL,
29     first_name         VARCHAR(100),
30     last_name          VARCHAR(100),
31     role               user_role      NOT NULL DEFAULT 'CLIENT',
32     enabled            BOOLEAN        NOT NULL DEFAULT TRUE,
33     created_at         TIMESTAMPTZ    NOT NULL DEFAULT now(),
34     updated_at         TIMESTAMPTZ    NOT NULL DEFAULT now(),
35
36     CONSTRAINT uq_app_user_email UNIQUE (email),
37     CONSTRAINT ck_app_user_email_format CHECK (email ~ '^[^@[:space:]]+@[^@[:space:]]+\.[^@[:space:]]+$'),
38     CONSTRAINT ck_app_user_email_lower CHECK (email = lower(email))
39 );
40
41 COMMENT ON TABLE app_user IS 'Comptes applicatifs, tous rôles confondus';
42 COMMENT ON COLUMN app_user.password_hash IS 'Hachage BCrypt – jamais le mot de passe en clair (corrige D-05)';
43 COMMENT ON COLUMN app_user.enabled IS 'Désactivation logique : on ne supprime pas un compte porteur d''historique';
44
45 CREATE TABLE customer (
46     id                BIGINT          GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
47     company_name       VARCHAR(255) NOT NULL,
48     siret              VARCHAR(14),
49     contact_email       VARCHAR(255),
50     contact_phone       VARCHAR(30),
51     billing_street      VARCHAR(255),
52     billing_post_code   VARCHAR(16),
53     billing_city        VARCHAR(100),
54     billing_country     CHAR(2)       NOT NULL DEFAULT 'FR',
55     active              BOOLEAN        NOT NULL DEFAULT TRUE,

```

```

56     created_at          TIMESTAMPTZ NOT NULL DEFAULT now(),
57     updated_at          TIMESTAMPTZ NOT NULL DEFAULT now(),
58
59     CONSTRAINT uq_customer_siret CHECK (siret IS NULL OR siret ~ '^[0-9]{14}$'),
60     CONSTRAINT ck_customer_country CHECK (billing_country ~ '^[A-Z]{2}$')
61 );
62
63 CREATE UNIQUE INDEX uq_customer_siret_idx ON customer (siret) WHERE siret IS NOT NULL;
64
65 COMMENT ON TABLE customer IS 'Entreprise cliente - distincte des comptes de connexion qui lui
sont rattachés';
66
67 CREATE TABLE customer_account (
68     customer_id BIGINT NOT NULL,
69     user_id      BIGINT NOT NULL,
70     created_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
71
72     CONSTRAINT pk_customer_account PRIMARY KEY (customer_id, user_id),
73     CONSTRAINT fk_customer_account_customer
74         FOREIGN KEY (customer_id) REFERENCES customer (id) ON DELETE CASCADE,
75     CONSTRAINT fk_customer_account_user
76         FOREIGN KEY (user_id) REFERENCES app_user (id) ON DELETE CASCADE
77 );
78
79 CREATE INDEX idx_customer_account_user ON customer_account (user_id);
80
81 CREATE TABLE delivery (
82     id                BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
83     reference          VARCHAR(20) NOT NULL,
84     customer_id        BIGINT NOT NULL,
85     created_by_user_id BIGINT,
86     status              delivery_status NOT NULL DEFAULT 'PENDING',
87
88     volume_m3          NUMERIC(10,3) NOT NULL,
89     weight_kg          NUMERIC(10,3) NOT NULL,
90
91     pickup_street      VARCHAR(255),
92     pickup_post_code   VARCHAR(16),
93     pickup_city         VARCHAR(100),
94     delivery_street    VARCHAR(255),
95     delivery_post_code VARCHAR(16),
96     delivery_city      VARCHAR(100),
97
98     requested_pickup_at TIMESTAMPTZ,
99     delivered_at        TIMESTAMPTZ,
100    price_ht            NUMERIC(12,2),
101
102    created_at          TIMESTAMPTZ NOT NULL DEFAULT now(),
103    updated_at          TIMESTAMPTZ NOT NULL DEFAULT now(),
104
105    CONSTRAINT uq_delivery_reference UNIQUE (reference),
106    CONSTRAINT fk_delivery_customer
107        FOREIGN KEY (customer_id) REFERENCES customer (id) ON DELETE RESTRICT,
108    CONSTRAINT fk_delivery_created_by
109        FOREIGN KEY (created_by_user_id) REFERENCES app_user (id) ON DELETE SET NULL,
110
111    CONSTRAINT ck_delivery_volume_positive CHECK (volume_m3 > 0),
112    CONSTRAINT ck_delivery_weight_positive CHECK (weight_kg > 0),
113    CONSTRAINT ck_delivery_price_positive CHECK (price_ht IS NULL OR price_ht >= 0),
114
115    CONSTRAINT ck_delivery_billed_has_price
116        CHECK (status <> 'BILLED' OR price_ht IS NOT NULL),
117    CONSTRAINT ck_delivery_delivered_has_date
118        CHECK (status <> 'DELIVERED' OR delivered_at IS NOT NULL)
119 );
120

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121 CREATE INDEX idx_delivery_customer_created ON delivery (customer_id, created_at DESC);
122 CREATE INDEX idx_delivery_status_created ON delivery (status, created_at DESC);
123 CREATE INDEX idx_delivery_pending ON delivery (created_at)
124 WHERE status = 'PENDING';
125
126 COMMENT ON TABLE delivery IS 'Commande de livraison - cycle PENDING → ACCEPTED → IN_TRANSIT →
DELIVERED → BILLED';
127 COMMENT ON COLUMN delivery.reference IS 'Référence métier lisible (LIV-2026-000123), communiquée
au client';
128 COMMENT ON COLUMN delivery.volume_m3 IS 'Volume en m³ - NUMERIC et non INT : un colis peut
mesurer 0,4 m³';
129
130 CREATE TABLE invoice (
131     id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
132     reference VARCHAR(20) NOT NULL,
133     customer_id BIGINT NOT NULL,
134     status invoice_status NOT NULL DEFAULT 'DRAFT',
135
136     issued_at TIMESTAMPTZ,
137     due_at TIMESTAMPTZ,
138     paid_at TIMESTAMPTZ,
139
140     total_ht NUMERIC(12,2) NOT NULL DEFAULT 0,
141     vat_rate NUMERIC(5,2) NOT NULL DEFAULT 20.00,
142     total_ttc NUMERIC(12,2) NOT NULL DEFAULT 0,
143
144     issued_by_user_id BIGINT,
145     created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
146     updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
147
148     CONSTRAINT uq_invoice_reference UNIQUE (reference),
149     CONSTRAINT fk_invoice_customer
150         FOREIGN KEY (customer_id) REFERENCES customer (id) ON DELETE RESTRICT,
151     CONSTRAINT fk_invoice_issued_by
152         FOREIGN KEY (issued_by_user_id) REFERENCES app_user (id) ON DELETE SET NULL,
153
154     CONSTRAINT ck_invoice_totals_positive CHECK (total_ht >= 0 AND total_ttc >= 0),
155     CONSTRAINT ck_invoice_vat_rate CHECK (vat_rate >= 0 AND vat_rate <= 100),
156     CONSTRAINT ck_invoice_issued_has_date
157         CHECK (status = 'DRAFT' OR issued_at IS NOT NULL)
158 );
159
160 CREATE INDEX idx_invoice_customer_issued ON invoice (customer_id, issued_at DESC);
161 CREATE INDEX idx_invoice_status ON invoice (status);
162
163 CREATE TABLE invoice_line (
164     id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
165     invoice_id BIGINT NOT NULL,
166     label VARCHAR(255) NOT NULL,
167     amount_ht NUMERIC(12,2) NOT NULL,
168
169     CONSTRAINT fk_invoice_line_invoice
170         FOREIGN KEY (invoice_id) REFERENCES invoice (id) ON DELETE CASCADE,
171
172     CONSTRAINT ck_invoice_line_amount CHECK (amount_ht >= 0)
173 );
174
175 CREATE INDEX idx_invoice_line_invoice ON invoice_line (invoice_id);
176
177 COMMENT ON COLUMN invoice_line.label IS
178     'Désignation de la prestation facturée, référence de livraison comprise - texte libre, non
contrôlé par le SGBD';
179
180 CREATE TABLE delivery_status_history (
181     id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
182     delivery_id BIGINT NOT NULL,

```

```

183     previous_status delivery_status,
184     new_status      delivery_status NOT NULL,
185     changed_by_user_id BIGINT,
186     comment         VARCHAR(500),
187     changed_at      TIMESTAMPTZ      NOT NULL DEFAULT now(),
188
189     CONSTRAINT fk_dsh_delivery
190         FOREIGN KEY (delivery_id) REFERENCES delivery (id) ON DELETE CASCADE,
191     CONSTRAINT fk_dsh_user
192         FOREIGN KEY (changed_by_user_id) REFERENCES app_user (id) ON DELETE SET NULL
193 );
194
195 CREATE INDEX idx_dsh_delivery_changed ON delivery_status_history (delivery_id, changed_at DESC);
196
197 COMMENT ON TABLE delivery_status_history IS 'Journal des transitions de statut – traçabilité
absente de l''application existante';
198
199 CREATE OR REPLACE FUNCTION set_updated_at() RETURNS TRIGGER AS $$
200 BEGIN
201     NEW.updated_at = now();
202     RETURN NEW;
203 END;
204 $$ LANGUAGE plpgsql;
205
206 CREATE TRIGGER trg_app_user_updated_at BEFORE UPDATE ON app_user
207     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
208 CREATE TRIGGER trg_customer_updated_at BEFORE UPDATE ON customer
209     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
210 CREATE TRIGGER trg_delivery_updated_at BEFORE UPDATE ON delivery
211     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
212 CREATE TRIGGER trg_invoice_updated_at BEFORE UPDATE ON invoice
213     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
214
215 CREATE SEQUENCE seq_delivery_reference START 1;
216 CREATE SEQUENCE seq_invoice_reference START 1;

```

02-seed.sql

Jeu de données de démonstration — développement et recette uniquement.

db/postgres/02-seed.sql Hors production 101 lignes

```

1  INSERT INTO app_user (email, password_hash, first_name, last_name, role) VALUES
2      ('admin@livrai.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Admin', 'LiVrai', 'ADMIN'),
3      ('meilin@livrai.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Meilin', 'Zhao', 'COMMERCIAL'),
4      ('commercial@livrai.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Paul', 'Mercier', 'COMMERCIAL'),
5      ('exploit@livrai.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Sarah', 'Bouvier', 'EXPLOITATION');
6
7  INSERT INTO customer (company_name, siret, contact_email, contact_phone,
8                      billing_street, billing_post_code, billing_city) VALUES
9      ('Boulangerie Durand', '12345678901234', 'contact@durand.fr', '0140000001', '12 rue du
Pain', '75011', 'Paris'),
10     ('Ateliers Moreau', '23456789012345', 'logistique@moreau.fr', '0140000002', '8 avenue des
Forges', '69003', 'Lyon'),
11     ('Pharmacie Centrale', NULL, 'gerant@pharma-c.fr', '0140000003', '3 place de la
Mairie', '33000', 'Bordeaux');
12
13  INSERT INTO app_user (email, password_hash, first_name, last_name, role) VALUES
14     ('gerant@durand.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Claire', 'Durand', 'CLIENT'),
15     ('logistique@durand.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Yanis', 'Berger', 'CLIENT'),
16     ('contact@moreau.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Hélène', 'Moreau', 'CLIENT'),
17     ('gerant@pharma-c.fr', '$2a$10$N9qo8uL0ickgx2ZMRZoMyeIjZAgcfl7p92ldGxad68LJZdL17lhWy',
'Marc', 'Lefèvre', 'CLIENT');
18
19  INSERT INTO customer_account (customer_id, user_id)
20  SELECT c.id, u.id FROM customer c, app_user u
21  WHERE (c.company_name = 'Boulangerie Durand' AND u.email IN ('gerant@durand.fr',
'logistique@durand.fr'))
22         OR (c.company_name = 'Ateliers Moreau' AND u.email = 'contact@moreau.fr')
23         OR (c.company_name = 'Pharmacie Centrale' AND u.email = 'gerant@pharma-c.fr');
24
25  INSERT INTO delivery (reference, customer_id, created_by_user_id, status,
26                      volume_m3, weight_kg,
27                      pickup_street, pickup_post_code, pickup_city,
28                      delivery_street, delivery_post_code, delivery_city,
29                      requested_pickup_at, delivered_at, price_ht)
30  SELECT
31      v.reference, c.id, u.id, v.status::delivery_status,
32      v.volume, v.weight,
33      v.p_street, v.p_cp, v.p_city,
34      v.d_street, v.d_cp, v.d_city,
35      v.pickup_at, v.delivered_at, v.price
36  FROM (VALUES
37      ('LIV-2026-000001', 'Boulangerie Durand', 'gerant@durand.fr', 'BILLED',
38       2.500, 120.000, '12 rue du Pain', '75011', 'Paris',
39       '45 rue de Rivoli', '75004', 'Paris',
40       now() - interval '20 days', now() - interval '18 days', 180.00),
41
42      ('LIV-2026-000002', 'Boulangerie Durand', 'logistique@durand.fr', 'DELIVERED',
43       1.200, 60.000, '12 rue du Pain', '75011', 'Paris',
44       '9 boulevard Voltaire', '75011', 'Paris',
45       now() - interval '6 days', now() - interval '4 days', 95.00),
46

```

```

47 ('LIV-2026-000003', 'Ateliers Moreau', 'contact@moreau.fr', 'IN_TRANSIT',
48 8.000, 1450.000, '8 avenue des Forges', '69003', 'Lyon',
49 '120 route de Vienne', '69007', 'Lyon',
50 now() - interval '1 day', NULL, 420.00),
51
52 ('LIV-2026-000004', 'Ateliers Moreau', 'contact@moreau.fr', 'ACCEPTED',
53 3.750, 300.000, '8 avenue des Forges', '69003', 'Lyon',
54 '5 quai Perrache', '69002', 'Lyon',
55 now() + interval '2 days', NULL, NULL),
56
57 ('LIV-2026-000005', 'Pharmacie Centrale', 'gerant@pharma-c.fr', 'PENDING',
58 0.400, 18.500, '3 place de la Mairie', '33000', 'Bordeaux',
59 '27 cours de l'Intendance', '33000', 'Bordeaux',
60 now() + interval '3 days', NULL, NULL),
61
62 ('LIV-2026-000006', 'Pharmacie Centrale', 'gerant@pharma-c.fr', 'REJECTED',
63 15.000, 3200.000, '3 place de la Mairie', '33000', 'Bordeaux',
64 '1 rue Sainte-Catherine', '33000', 'Bordeaux',
65 now() - interval '2 days', NULL, NULL)
66 ) AS v(reference, company, creator_email, status, volume, weight,
67 p_street, p_cp, p_city, d_street, d_cp, d_city,
68 pickup_at, delivered_at, price)
69 JOIN customer c ON c.company_name = v.company
70 JOIN app_user u ON u.email = v.creator_email;
71
72 INSERT INTO delivery_status_history (delivery_id, previous_status, new_status,
73 changed_by_user_id, comment, changed_at)
74 SELECT d.id, h.prev::delivery_status, h.next::delivery_status, u.id, h.comment, h.at
75 FROM (VALUES
76 ('LIV-2026-000001', NULL, 'PENDING', 'gerant@durand.fr', 'Commande créée par le
client', now() - interval '22 days'),
77 ('LIV-2026-000001', 'PENDING', 'ACCEPTED', 'exploit@livrai.fr', 'Créneau confirmé',
now() - interval '21 days'),
78 ('LIV-2026-000001', 'ACCEPTED', 'IN_TRANSIT', 'exploit@livrai.fr', 'Enlèvement effectué',
now() - interval '20 days'),
79 ('LIV-2026-000001', 'IN_TRANSIT', 'DELIVERED', 'exploit@livrai.fr', 'Remise au
destinataire', now() - interval '18 days'),
80 ('LIV-2026-000001', 'DELIVERED', 'BILLED', 'meilin@livrai.fr', 'Facture FAC-2026-000001
émise', now() - interval '15 days'),
81 ('LIV-2026-000006', NULL, 'PENDING', 'gerant@pharma-c.fr', 'Commande créée par le
client', now() - interval '3 days'),
82 ('LIV-2026-000006', 'PENDING', 'REJECTED', 'exploit@livrai.fr', 'Poids au-delà de la
capacité', now() - interval '2 days')
83 ) AS h(reference, prev, next, actor_email, comment, at)
84 JOIN delivery d ON d.reference = h.reference
85 JOIN app_user u ON u.email = h.actor_email;
86
87 INSERT INTO invoice (reference, customer_id, status, issued_at, due_at,
88 total_ht, vat_rate, total_ttc, issued_by_user_id)
89 SELECT 'FAC-2026-000001', c.id, 'ISSUED',
90 now() - interval '15 days', now() + interval '15 days',
91 180.00, 20.00, 216.00, u.id
92 FROM customer c, app_user u
93 WHERE c.company_name = 'Boulangerie Durand' AND u.email = 'meilin@livrai.fr';
94
95 INSERT INTO invoice_line (invoice_id, label, amount_ht)
96 SELECT i.id, 'Livraison LIV-2026-000001 - Paris 11e → Paris 4e', 180.00
97 FROM invoice i
98 WHERE i.reference = 'FAC-2026-000001';
99
100 SELECT setval('seq_delivery_reference', 6);
101 SELECT setval('seq_invoice_reference', 1);

```

Vérification. Les deux scripts ont été exécutés sur une instance PostgreSQL 18 en conteneur. Le schéma crée sept tables, trois types énumérés et neuf index. Les contraintes ont été éprouvées par insertion volontaire de valeurs invalides : un volume de `-99` est rejeté par `ck_delivery_volume_positive`.

Annexe au document d'architecture du CRM LiVrai — septembre 2026.